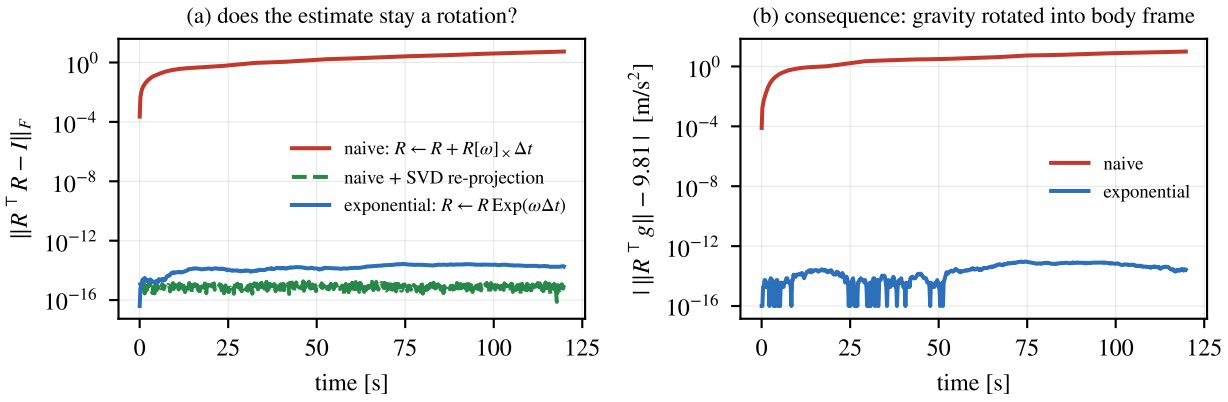
(a) does the estimate stay a rotation?
$\|R^\top R - I\|_F$
10^0
10^{-4}
10^{-8}
10^{-12}
10^{-16}
naive: $R \leftarrow R + R[\omega]_\times \Delta t$
naive + SVD re-projection
exponential: $R \leftarrow R\,\mathrm{Exp}(\omega \Delta t)$
0
25
50
75
100
125
time [s]
(b) consequence: gravity rotated into body frame
$|\,\|R^\top g\| - 9.811\,|$ [m/s^2]
10^0
10^{-4}
10^{-8}
10^{-12}
10^{-16}
naive
exponential
0
25
50
75
100
125
time [s]